Wildfire Risk Classification Model

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Abstract

Wildfire risk in Canada is escalating due to climate change, with the 2023 season marking the worst on record, affecting 15 million hectares and displacing 232,000 people. Northern Alberta, particularly Mackenzie County, faces increasing fire frequency and intensity, yet remains understudied. This research addresses critical gaps in wildfire prediction through an integrated machine learning approach. Using historical fire intensity data from 2014 to 2024, combined with environmental variables including climate conditions, topography, land use/land cover, and human infrastructure, developed a Random Forest classifier to predict future wildfire intensity across six categories: nil, low, moderate, high, very high, and extreme. Data extraction required developing a Python-based automated system to process interactive fire danger maps from Natural Resources Canada. Following data preprocessing, normalization, and integration with climate, topographic, and infrastructure layers, the model was trained and validated using the scikit-learn library. The Random Forest model achieved 96% overall accuracy, with particularly strong performance in identifying critical high-risk areas, achieving precision and recall values of 0.96-0.98 and 0.95-0.97 for "very high" and "extreme" categories respectively. The confusion matrix demonstrated strong diagonal dominance, indicating accurate classification across all intensity levels with minimal misclassification primarily between adjacent categories. Model outputs include spatially and temporally linked predicted and validated fire intensity maps, enabling day-by-day comparison of predicted versus actual fire conditions. The predictive capability demonstrated through examples from June and July 2022 shows strong alignment between predicted and true fire intensity patterns. This research provides emergency management agencies with a robust tool for forecasting wildfire behavior, enabling data-driven resource allocation and proactive risk mitigation planning in fire-prone regions of Northern Alberta and beyond.

Keywords: wildfire prediction, machine learning, Random Forest, fire intensity, GIS, Northern Alberta, climate change, risk assessment

1. Introduction

Canada is at risk of major yearly wildfire danger, with climate change expected to worsen the frequency and intensity of wildfires (Ahmed & Hassan, 2023; Whitman et al., 2022; Johnson et al., 2020). In 2023, Canada witnessed its worst wildfire season, with 15 million hectares destroyed and 232,000 persons affected (Jain et al., 2024). Wildfires cause severe air quality concerns and interruptions, significantly impacting First Nation land (Boulanger et al., 2024). The catastrophic event similar to those in 2023 are predicted to occur 6.3 to 10.8 times more often by the end of the century (Jones et al., 2024). To meet these problems, cutting-edge fire control and prevention measures are required (Jain et al., 2024).

Despite the limited scope fire risk research in Canada (Johnson et al., 2020), there are fire risk studies utilizing GIS happening on global scale (Zacharakis & Tsihrintzis, 2023). There are studies using multi-criteria analysis (MCA) methods to build fire risk zone maps (Cetin et al., 2022; Sivirkaya et al., 2022; Parajuli et al., 2020; Coban & Erdin, 2020) and to model forest fire hazards (Faramarzi et al., 2021). Although, there are examples of fire research in Canada (Erni et al., 2024; Zacharakis & Tsihrintzis, 2023), including using multi-criteria analysis in Ontario (Bennett et al., 2020) and using statistical analysis in Alberta (Whitman et al., 2022; Beverly & McLoughlin, 2019). Additionally, the prediction of fire has transitioned from quantifying wildfire with GIS techniques to machine learning (ML) globally (Zacharakis & Tsihrintzis, 2023), with examples for assessing fire activity conducted in the United States (Finney et al., 2011; Malik et al., 2021) and Korea (Piao et al., 2022).

Moreover, to improve fire preparedness and response, the planning of evacuation routes is essential. There have been GIS methods used to help plan evacuation routes for wildfires, globally, these include using network analysis (Wang et al., 2014; Yi et al., 2012) and modeling (Li et al., 2019). Research in effective evacuation planning enhances decision making and formulates future data collection on critical wildfire evacuation challenges (Essen et al., 2022).

The study aims to address the research gaps in fire prediction, preparedness, and response in Canada, as well as the gaps in fire risk validation, and evacuation route identification in Northern Alberta by using Mackenzie County as a case study. In 2012, Mackenzie County faced challenges during evacuation process due to the absence of details regarding the surrounding wildfire and smoke and general escape strategy, hindered and complicated the evacuation of the Dene Tha' community of Zama City (Mottershead et al., 2020). Limited research has been conducted in the Mackenzie County region of Alberta, which raises significant concerns for the various residents and First Nation communities in the area.

The project addresses these knowledge gaps through the research question: How can GIS-based analysis be used to evaluate wildfire risk areas, model future wildfire activity, and plan evacuation routes in Mackenzie County, Alberta?

First, we aim to identify areas within Mackenzie County that are at high risk of future wildfires. Using multi-criteria analysis (MCA), we will develop a validated fire risk map that highlights high-risk regions by integrating climate, topographic, land cover, and human infrastructure variables. The map's accuracy will be ensured through validation against historical wildfire occurrence data, providing a reliable foundation for understanding spatial patterns of wildfire susceptibility across the county.

Second, we will develop a machine learning model capable of predicting future wildfire behavior in high-risk regions. This model will analyze historical fire intensity data alongside environmental variables to forecast potential fire activity, generating predictions of spatial patterns and intensity levels over time. By leveraging patterns learned from past fire events, this predictive capability will provide emergency management agencies with valuable insights to support proactive wildfire management, resource allocation, and strategic planning.

Third, we will determine the most effective evacuation routes for communities located in high-risk zones. Through network analysis of existing road infrastructure and settlement locations, we will identify optimal evacuation paths that ensure residents have safe and efficient routes during fire emergencies. This analysis will evaluate multiple fire scenarios to assess evacuation feasibility and identify potential infrastructure vulnerabilities that could impede successful community evacuation.

Collectively, these three objectives provide an integrated approach to enhancing wildfire risk assessment, prediction, and emergency response strategies in Mackenzie County. By combining spatial risk mapping, temporal prediction modeling, and evacuation route planning, this research delivers actionable tools and insights that support evidence-based decision-making for wildfire preparedness and community safety in Northern Alberta.

2. Data overview

This study integrates geospatial and environmental datasets to support machine learning, based wildfire intensity prediction in Mackenzie County, Alberta. Key inputs include historical fire intensity and hotspot data from Natural Resources Canada, climate variables (precipitation, temperature, and wind), topographic factors (slope, aspect, elevation), land cover data, and human infrastructure layers such as roads and highways. These datasets (Table 1) capture the environmental and anthropogenic drivers influencing fire behavior and spread. All data were preprocessed in QGIS and ArcGIS, clipped to the study area, standardized to a 100-meter raster resolution, normalized, and reprojected to a common coordinate system to ensure spatial consistency. The resulting integrated dataset provides the foundation for training a Random Forest classifier to predict spatial and temporal wildfire intensity patterns.

Table 1

Data sources of the project

Data	Source
Wind Speed	Global Wind Atlas
Fire Intensity	Natural Resources Canada
Fire Hotspots: 2009 to 2024	Natural Resources Canada

3. Modeling Future Wildfire Scenarios

The project uses a machine learning algorithm to develop a predictive model that forecasts future wildfire risk areas by analyzing historical fire occurrences and intensity. For data analysis, a random forest classifier from the Scikit Learn Python library was applied. For wildfire prediction, researchers discovered that random forest model is considerably more accurate than other machine learning algorithms (Singh & Jeganathan, 2024; Goa et al, 2023). Scikit Learn is a well-known open-source machine learning package that offers simple tools for predictive data analysis, and it is built on libraries such as NumPy, SciPy, and matplotlib. The random forest algorithm produces decision trees from various training sets and averages their predictions to form a single output (Shahdeo et al., 2020). This method enables the model to learn connections among different features using the provided training data. The Scikit Learn classifier employs an optimal split strategy, is proficient at handling datasets with multiple features, and effectively manages extraneous variables. The algorithm demonstrates a high level of accuracy in matching the original data with the validated data, making it an ideal algorithm for model development (Iban & Sekertekin, 2022; Collins et al., 2018).

3.1 Data Cleaning and Pre-Processing

The model incorporates several datasets, including climate conditions, topography, land use/land cover, human infrastructure, fire hotspots, and fire intensity. Most datasets for model development align with MCA data, with additional information on wind speed, fire hotspots from 2009 to 2024, and fire intensity included. For a complete list of datasets used, refer to Table 6.

QGIS and ArcGIS are the foundation for dataset cleaning. The initial step involved clipping the datasets to Mackenzie County using the clipping geoprocessing tool in QGIS. After clipping, the vector data is transformed into raster format using the rasterize conversion tool, where the input is the vector layer, and the burn-in value corresponds to the Forest Fire Weather

Index (FWI), the raster size units are set as pixels, with a vertical and horizontal resolution of 100, keeping other parameters at their defaults.

Table 2

Datasets used for machine learning model development

Category	Datasets
Climate	Precipitation
	Temperature
	Wind Speed
Topography	Slope
	Aspect
	Elevation
Landcover	Landcover/Land use
Fire Hotspots	Hotspots
Fire Intensity	Intensity
Human Infrastructure	Provincial Highways
	Junctions
	Blocked Passages
	Road Network

To ensure consistency in scale, all raster data are normalized to a range of 0 to 1 using raster calculator tool in QGIS. The formula used is illustrated in Figure 1. Additionally, ArcGIS is employed to adjust the projection to the NAD 1983 10TM AEP Forest which is relevant to the study area.

$$\text{Normalized} = \frac{\text{Raster} - \min(\text{Raster})}{\max(\text{Raster}) - \min(\text{Raster})}$$

Figure 1

Formula used to normalize the datasets

Vector data, including fire hotspots and human infrastructure, are overlaid onto the normalized raster datasets (covering climate, topography, and land use/land cover) to extract CSV values for each overlaying vector layer, thereby generating a detailed dataset for the model. The sample raster values tool under raster analysis extracts CSV table sample values with vector data as the input layers, and raster layers, while setting the output column to "RVALUES". The output was a

comprehensive CSV dataset with attributes including latitude, longitude, date, corresponding raster values and other associated vector attribute. This process was repeated for fire hotspots from 2009 to 2024 and other vector datasets, including human infrastructure. This approach ensured that the datasets were accurately preprocessed and formatted for training the machine learning model, providing a reliable foundation for wildfire risk map generation.

3.2 Extracting Fire Intensity Data

To build the wildfire prediction model, we needed historical fire intensity data spanning multiple years. Unfortunately, no open-source datasets were readily available for download. The only accessible source was an interactive map from Natural Resources Canada (NRCAN) that displayed fire danger levels categorized as nil, low, moderate, high, very high, and extreme, each represented by a specific color. To overcome this challenge, we developed a systematic process to extract, process, and prepare the data for use.

The first step was to automate the downloading of these maps. By analyzing the URLs of the maps, we identified a consistent pattern based on the date. Using this information, we wrote a script to generate and access URLs for all the available dates between 2014 and 2024, allowing us to efficiently download the entire collection of maps.

Once the maps were downloaded, the next step involved extracting the fire intensity data from the images. For each pixel in the map, we identified its RGB color and matched it to the closest predefined intensity level. This process converted the visual map data into labeled fire intensity values, which were saved as structured datasets, including geographic coordinates, dates, and intensity labels.

The extracted data needed further refinement to align with our study area, Mackenzie County. We processed the datasets to correct any geographic misalignments, apply necessary transformations, and clip the data to the county's boundaries. This ensured that the data accurately represented fire intensities specific to the region of interest.

Finally, the individual datasets from all the years were merged into a single comprehensive fire intensity dataset. This step combined data across different dates into one unified file, making it easier to integrate with other datasets like elevation, precipitation, and landcover. The resulting dataset provided a complete and consistent representation of fire intensities, ready for use in the machine learning model alongside other environmental and climatic variables.

3.3 Dataset Merge for Model

To create a training dataset for the machine learning model, it was necessary to combine the fire intensity data with the other datasets collected during the earlier data preparation process. These datasets included a variety of spatial and environmental features, all pre-processed and aligned as described in the data cleaning and pre-processing phase.

The merging process began with selecting the fire intensity dataset as the base file. Using its geographic coordinates, other datasets were matched by identifying the nearest corresponding data points. This ensured that each location in the fire intensity dataset was paired with additional attributes from the other datasets, creating a complete profile for each point. To handle the spatial matching, a nearest-neighbor approach was applied. This allowed for efficient pairing of data points across all datasets, ensuring accuracy while minimizing processing time. Once all datasets were aligned, the merged file contained all the key variables needed for the machine learning model.

The final output was a comprehensive benchmark dataset that integrated fire intensity information with a range of relevant environmental and climatic features. This combined dataset served as the foundation for training the model, enabling it to analyze multiple factors simultaneously and make accurate predictions about wildfire behavior.

The evaluation results, including the predicted and actual fire intensity labels, were saved along with their corresponding geographic coordinates (latitude and longitude) and temporal information (year, month, and day). This ensured that each prediction could be linked to a specific point in space and time for detailed analysis. Additionally, shapefiles were created for both the true and predicted labels, enabling visualization of the differences for each day. This spatial representation allowed us to assess how well the model performed across the study area and provided insights for refining the model further.

3.4 Training and Validation

The training and validation process was carried out using a Python script built with the scikit-learn machine learning library. This script automated key steps, from data preparation to model evaluation, ensuring consistency and efficiency.

The dataset was divided into training and testing subsets to allow the model to learn from one portion of the data and be evaluated on a separate, unseen portion. Features, such as spatial and environmental attributes, were standardized using the *StandardScaler* function to ensure uniform scaling, which is critical for optimizing the performance of machine learning algorithms. The target labels, representing fire intensity levels, were encoded into numerical values using *LabelEncoder* for compatibility with the model.

A Random Forest classifier was chosen for its robustness and ability to handle complex relationships within the data. The *RandomForestClassifier* function from scikit-learn was used to

train the model, allowing it to learn patterns and associations between features and fire intensity levels. This algorithm was particularly suited to the task as it excels in handling both categorical and numerical data while maintaining high accuracy and reducing overfitting.

To evaluate the model, predictions were generated for the test subset, and the results were compared to the actual fire intensity labels. The script employed the *accuracy*_score, *classification_report*, and *confusion*_matrix functions from scikit-learn to provide a comprehensive evaluation of the model's performance. The *accuracy_score* measured the overall correctness of predictions, while the *classification_report* provided detailed metrics like precision, recall, and F1-score for each fire intensity category. The *confusion_matrix* offered insights into misclassifications, helping to identify areas where the model could improve.

Lastly, the model produces visualizations that can be used in assessing risk areas. The maps generated by the model are converted into readable shapefiles using ArcGIS. The symbology of these maps has been adjusted to create a uniform layer resembling the fire intensity map produced by NRCAN. The intensity scale of the map is designed to include low (blue), moderate (green), high (yellow), very high (orange), and extreme (red) levels, aligning with the original fire intensity maps.

The workflow diagram, Figure 2**,** illustrates the development process for the machine model, starting with data extraction, followed by data cleaning, and concluding with model validation.

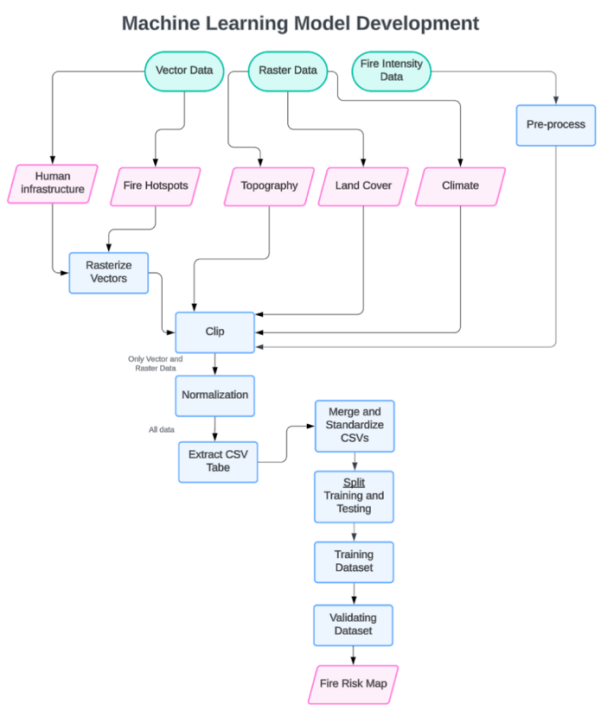

Figure 2

Machine learning model development workflow

4. Model Generated Wildfire Risk Visualization

The model's evaluation metrics provide a detailed understanding of its performance in predicting wildfire intensities. With an overall accuracy of 96%, the model demonstrates robust predictive capabilities, effectively identifying high-risk areas and minimizing misclassifications. This high accuracy indicates that the model has learned the relationships between the features and target labels well, making it reliable for practical applications in wildfire risk management.

The classification report (Figure 3) provides detailed metrics for all six fire intensity categories: nil, low, moderate, high, very high, and extreme. The model achieved precision and recall values of 0.93 and 0.91, respectively, for "nil," demonstrating strong performance in recognizing areas with no fire intensity. For "low," the precision was 0.89, and recall was 0.88, indicating slight room for improvement. "Moderate" intensity showed balanced results with a precision of 0.92 and recall of 0.90. The "high" category performed similarly, with a precision of 0.93 and recall of 0.92. The "very high" and "extreme" categories excelled, achieving precision of 0.96 and 0.98 and recall of 0.95 and 0.97, respectively. The weighted average F1-score of 0.96 underscores the model's strong overall performance, ensuring reliability across all intensity levels.

Classification Report:

	precision	recall	f1-score	support
0	0.91	0.78	0.84	2265
1	0.90	0.92	0.91	25531
2	0.95	0.95	0.95	21201
3	0.90	0.88	0.89	18055
4	1.00	1.00	1.00	83014
5	0.91	0.91	0.91	17146
accuracy			0.96	167212
macro avg	0.93	0.91	0.92	167212
weighted avg	0.96	0.96	0.96	167212

Figure 3

Classification Report Results

The confusion matrix (Figure 4) offers a clear visualization of how well the model predicts each fire intensity category. The goal of the confusion matrix is for the "diagonal" values—representing correctly classified instances—to be the highest. This is indeed what we observe, with most predictions aligning with the true labels. Misclassifications are minimal and generally occur between adjacent categories, such as "low" and "moderate," which is expected given the similarities between these intensities. This high diagonal dominance confirms the model's ability to accurately classify fire intensity levels while leaving room for improvement in borderline cases.

```
Confusion Matrix:
 [[ 1759    18     2     1     0   485]
 [   10 23428    75  1001     5  1012]
 [    0    74 20243   774    93    17]
 [    4  1178   904 15871    40    58]
 [    0     7    58    22 82903    24]
 [  154  1302    11    49     4 15626]]
Accuracy: 0.9558524507810444
```

Figure 4

Confusion Matrix

The recall and precision graphs (Figure 5) visually emphasize the model's reliability in balancing false positives and false negatives. For categories like "very high" and "extreme," the graphs show nearly perfect performance, indicating that the model reliably captures critical high-risk areas. Meanwhile, categories like "nil" and "low" maintain strong recall and precision values, ensuring that less intense fire zones are also appropriately classified. These metrics together underscore the model's effectiveness in providing actionable wildfire predictions.

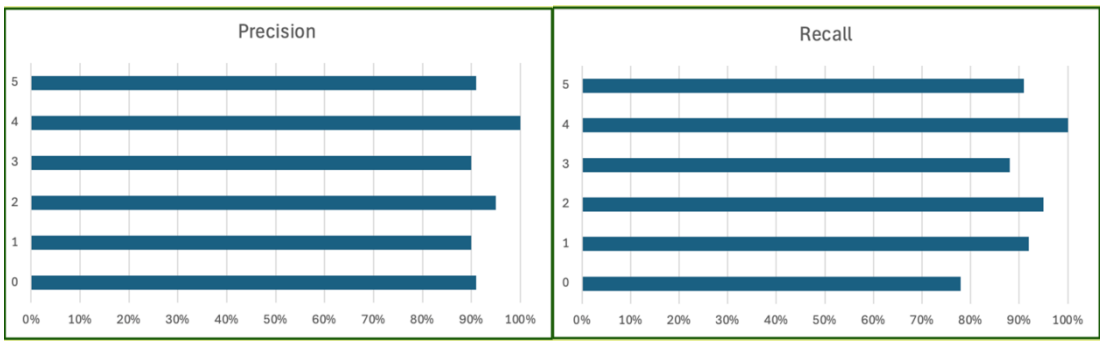

Figure 5

Precision and Recall Graphs

By combining these evaluation results, the model offers a robust tool for identifying and visualizing wildfire risk. The ability to link predicted and true labels spatially and temporally allows for the generation of maps that highlight differences in predictions for each day, making the model outputs interpretable and actionable for stakeholders. These insights are critical for advancing wildfire preparedness and response strategies.

The outcomes of the model consist of both validated and predicted fire intensity maps. For example, a map dated July 7, 2022, (Figure 6) illustrates the true fire intensity validated by the model alongside the predicted results. This map demonstrates a scale ranging from low to very high, with low intensity covering a substantial portion of the area. The true label corresponds to the original map from NRCAN, while the predicted label reflects data after it has

been trained and validated using 2022 data. These maps exemplify how the model can effectively predict fire intensity over time.

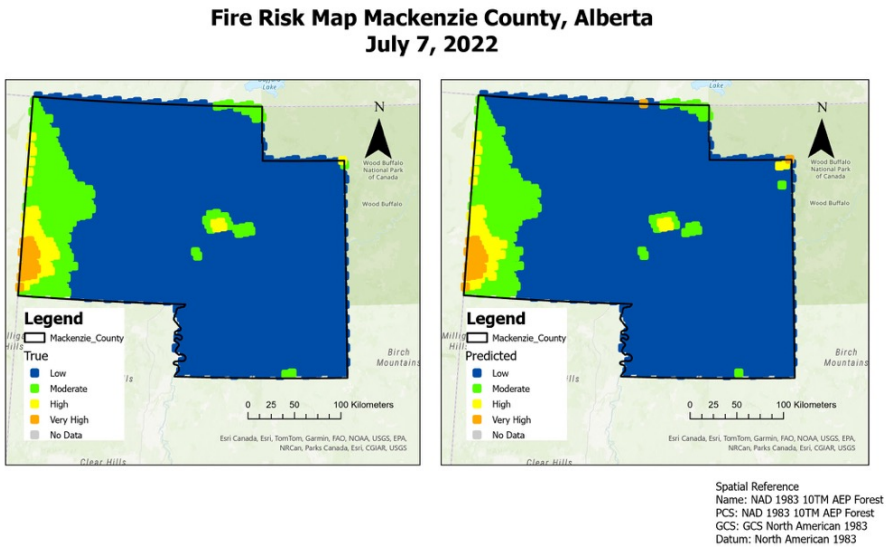

Figure 6

Illustrates true and predicted fire intensity maps of July 07, 2022

Another example presents a similar analysis of fire intensity for June 4, 2022 (Figure 7). On this day, the map indicates predominantly very high-risk and extreme areas. There is a significant alignment between the true and predicted labeled maps for each day depicted, suggesting that the model successfully captures the critical factors influencing fire intensity and provides accurate predictions.

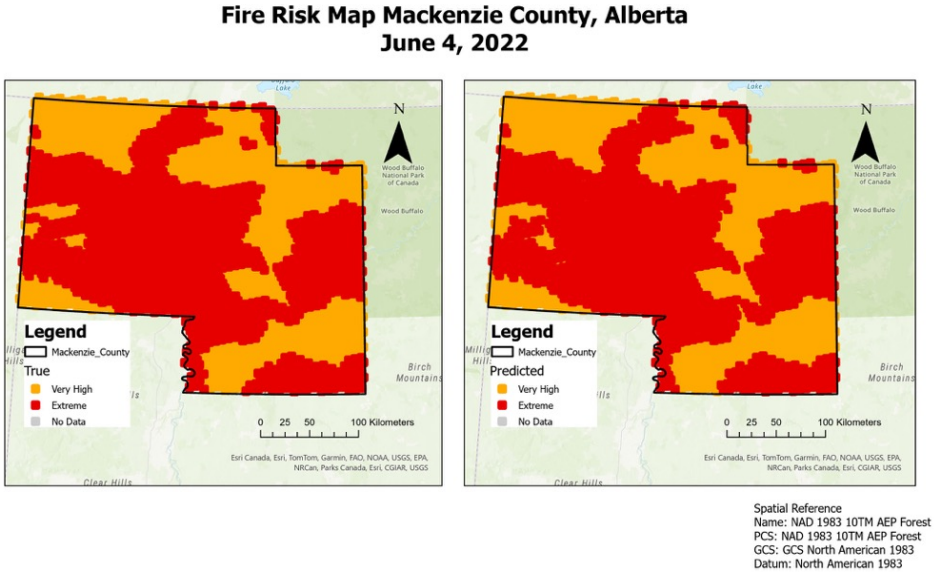

Figure 7

Illustrates true and predicted fire intensity map of June 04, 2022

5. Machine Learning Model Capabilities

The machine learning model demonstrates strong potential as a valuable tool for predicting and managing wildfire risks in Mackenzie County and beyond. It excels at identifying high-risk intensity levels like "very high" and "extreme," ensuring critical areas are flagged accurately. This capability is essential for effective emergency planning and resource allocation.

While the model's accuracy is impressive, it struggles with distinguishing lower intensity levels like "low" and "moderate," likely because these conditions are quite similar. Improving the model or adding more data could help it handle these cases better. The maps comparing predicted and actual intensities show where the model is doing well and where it could improve. High-risk areas match closely with predictions, while moderate-risk zones show some small differences, giving useful insights for making the model even better.

Forecasted weather data can improve the model, which has a 96% accuracy for predicting wildfire risk in an area of interest. This would make it a flexible tool for the decision-making process during wildfire outbreaks by enabling it to adjust to changing conditions and deliver real-time estimates. It could evolve into a robust wildfire management solution with future developments.

6. Limitation

This study faced several limitations that are important to consider when interpreting the results. One of the main challenges was accessing historical fire intensity data. Fire intensity information from Natural Resources Canada was only available through interactive web maps rather than downloadable datasets, which required developing a custom Python script to extract daily data from 2014 to 2024. This process was time-intensive and ultimately limited the analysis to a seven-year window. A longer historical record would likely improve the model's ability to capture rare extreme events, long-term climate patterns, and broader fire behavior trends. Future work would benefit greatly from standardized, machine-learning–ready fire intensity datasets to reduce preprocessing effort and enable deeper temporal analysis.

Although the Random Forest model achieved a high overall accuracy of 96%, performance varied by fire intensity category. The model performed very well in identifying high-risk conditions, with strong precision and recall for the "very high" and "extreme" classes. However, it was less effective at distinguishing between lower-intensity categories such as "low" and "moderate." This is likely due to the subtle environmental differences between these classes, where small variations in weather or fuel conditions are difficult for the model to separate. Incorporating additional variables such as fuel moisture, vegetation health indices, or lightning data could help improve classification in these borderline cases. In addition, the model relies

entirely on historical data and does not include real-time or forecasted weather inputs, which limits its usefulness for operational fire response during active fire seasons.

The model was developed specifically for Mackenzie County, and its results may not directly translate to regions with different climates, vegetation types, or fire regimes. While the overall approach can be applied elsewhere, model parameters and intensity thresholds are tuned to local boreal forest conditions. Validation was performed using NRCAN fire danger maps, which are themselves modeled products rather than direct fire measurements, introducing some uncertainty. Validation against independent fire occurrence or intensity data would strengthen confidence in the results. Finally, all variables were processed at a 100-meter spatial resolution, which may smooth out fine-scale terrain or vegetation features that influence localized fire behavior.

7. Conclusion

This research developed and validated a machine learning–based approach to predicting wildfire intensity in Mackenzie County, Alberta, helping address gaps in fire risk forecasting for Northern Canada. The Random Forest model achieved an overall accuracy of 96% across six fire intensity categories and performed especially well at identifying high-risk conditions. Its strong performance for the “very high” and “extreme” categories shows clear potential for supporting emergency preparedness, resource allocation, and evacuation planning.

A key contribution of this work was overcoming major data availability challenges. Because historical fire intensity data were only available through interactive web maps, a custom automated extraction process was developed to convert these maps into structured spatial datasets suitable for machine learning. Validation against observed fire intensity patterns for selected dates in 2022 showed strong spatial agreement between predicted and actual conditions, suggesting the model can provide practical, location-specific insights for proactive fire management.

There are also clear opportunities to build on this work. Expanding the training dataset to include longer historical records would help capture rare extreme events and longer-term climate trends. Incorporating additional variables such as fuel moisture, vegetation health indices, or higher-resolution data could improve the model’s ability to distinguish between lower-intensity fire categories. Most importantly, integrating real-time or forecasted weather data would move this model from a retrospective analysis tool toward an operational system capable of supporting day-to-day fire management decisions. While the overall framework can be applied to other regions, local calibration would be needed to reflect differences in vegetation, climate, and fire behavior.

Overall, the high accuracy achieved in this study represents a meaningful step forward in wildfire prediction for Northern Alberta. By providing clearer insight into where severe fire conditions are most likely to occur, this approach can help emergency managers, local

governments, and First Nations communities plan more effectively and respond proactively. As wildfire risk continues to increase under a changing climate, combining machine learning with comprehensive environmental data offers a practical and powerful way to support informed decision-making and protect communities and natural landscapes.

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